Omkar Najan

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Summary

Backend software engineer with 3+ years of experience owning systems end to end — from **HLD/LLD** through **production rollout** and on-call. Builds **distributed, event-driven services** in **Java and Spring Boot** on **AWS**, including a **zero-downtime cross-region migration** of a live order platform (**1.2K+ events/minute**) and a **centralised notification service** spanning 5 travel domains. **500+ DSA problems** solved; B.Tech, Computer Science (**CGPA 9.3/10**).

Skills

Languages	Java (21), C++, SQL, JavaScript Python
Backend & Distributed Systems	Spring Boot, Spring MVC, REST APIs, Microservices, Event-Driven Architecture, Idempotency, Fault Tolerance, Caching
Cloud & DevOps	AWS (Lambda, SNS, SQS, DynamoDB, ECS, API Gateway, S3, ElastiCache, CloudWatch, KMS, IAM), AWS CDK, Docker, CI/CD, Observability & On-Call Operations
Design, Data & AI	HLD/LLD, System Design, Feature-Flag Rollouts, DynamoDB, MySQL, Redis, LLM Integrations, Agentic Workflows

Experience

Software Engineer Jul 2025 – Jul 2026
Amazon *Hyderabad, India*
Java, Spring Boot, AWS, Distributed Systems, Event-Driven Architecture

- Owned the HLD, LLD, implementation and production rollout for re-architecting a production order-event service from EC2 (Java 8, Spring) to **AWS Lambda (Java 21, SnapStart)**, processing **1.2K+ events per minute**; built the event-driven pipeline on **SNS, SQS FIFO and DynamoDB** with DLQs, per-order ordering, deduplication, retries and **idempotent processing** — cutting cold-start latency from ∼**3 seconds to 500 ms**, with **95% unit-test coverage**.
- Led the **zero-downtime cross-region cutover**: 2 Lambda functions, 8 per-vertical FIFO queues and 2 DynamoDB tables converted to **Global Tables** (replication under 1 second), with **feature-flagged traffic shifting** (1% to 100%), 7 days of stable full traffic and **rollback in under 10 minutes**; SNS-based switching cut publisher latency from ∼**100 ms to 15 ms**.
- Owned the end-to-end design, build and production support of a **centralised communication service** (Spring Boot, ECS) replacing duplicated notification logic across **5 travel domains** — Email, SMS, WhatsApp and PDF e-tickets; **Factory/Strategy** channel routing, 14+ Mustache templates with **ElastiCache cache-aside** retrieval, idempotent sends, **KMS-encrypted PII** and full logging, metrics and tracing.
- Migrated the User Communication Service out of eu-west-1 under the org-wide **40% server-reduction** initiative, modernising legacy template-based infrastructure onto **AWS CDK**; designed the target architecture (API Gateway, load balancer, ECS, KMS-encrypted S3 and SNS), including **cross-region consistency** between e-ticket PDF storage and notification delivery, so a notification can never reference a PDF that is missing in its region.
- Contributed to the HLD for hotel cross-sell after flight, train and bus bookings (business case targeted **2,000 additional daily bookings**); designed its Transaction Details Page rendering (6,196 reads/hour) with a **Redis read-through cache** addressing a measured **1,471 ms average recommendation latency**, and scheduled WhatsApp notifications (max 2 per booking) with suppression rules and **at-most-once delivery**.
- Implemented the payment-service client migration to an **SNS-based integration** (reusable CDK constructs, feature-flagged rollout, cross-account IAM/STS, SigV4) and led **shared CDK library upgrades** adopted by 2 teams; contributed to the OTD/OTS order-translation services that surface bookings in Your Orders and Amazon Pay, including **dual-region queue consumption** during the migration.
- Built an **LLM-powered agent** that automates read-only on-call investigation across logs, metrics, alarms and tickets using cross-account CloudWatch, to **speed up root-cause analysis**; secured it with a **defence-in-depth safety model** (tool allowlist, command-deny hooks, read-only IAM, SELECT-only database access) packaged as Infrastructure as Code; resolved multiple **Sev2 incidents** on call.

Software Engineer Aug 2023 – Jun 2025
Cytel Inc. *Pune, India*
Java, Spring Boot, C++, Python, Docker, CI/CD

- Re-engineered the communication layer of a clinical-trial simulation product from XML to JSON, integrating a **C++ engine** with a **Java/Spring Boot** backend to enable a desktop-to-web migration; designed a modular interface allowing new simulation types to be added without altering the core engine.
- Developed **stateless Java REST APIs** (Spring Boot) orchestrating **15+ clinical simulations** against the C++ engine: requests are validated and executed **synchronously** over the JSON interface, with **no database in the request path**;

implemented **structured error mapping** to translate raw engine failures into meaningful API error states for the frontend.

- Built a secure internal **knowledge bot for Microsoft Teams** using **LangChain** and a **privately hosted Azure LLM**, enabling natural-language queries against Jira and Confluence while keeping data inside company infrastructure, reducing time spent answering repetitive questions.
- Replaced manual API testing with data-driven automated suites using the **Karate framework** (JavaScript), validating JSON simulation results against CSV input/output pairs and **cutting testing time by 60%**.
- Automated CI/CD using **Bitbucket pipelines** and **Python** to deploy Docker containers to VMs; reviewed code and mentored one intern.

DC Analyst Jan 2023 – Aug 2023
Deloitte USI *Hyderabad, India*
SQL, Excel

- Designed and executed **SQL workflows** to extract, clean and validate payroll data for **50,000+ employees**, ensuring accuracy and compliance with financial regulations.
- Identified and resolved **payroll discrepancies** through data validation checks and reporting, coordinating with US and India teams during reconciliation cycles.

Education

B.Tech., Computer Science and Engineering 2018 – 2022
MIT World Peace University, Pune *CGPA: **9.3/10***

Achievements

- Solved **500+ data structures and algorithms** problems on LeetCode and similar platforms.